

Creating a Page Layout with the Bootstrap Grid System



Gill Cleeren

CTO Xebia Microsoft Services Belgium

@gillcleeren | www.snowball.be



Understanding the Bootstrap Grid System



It All Started with a Table...

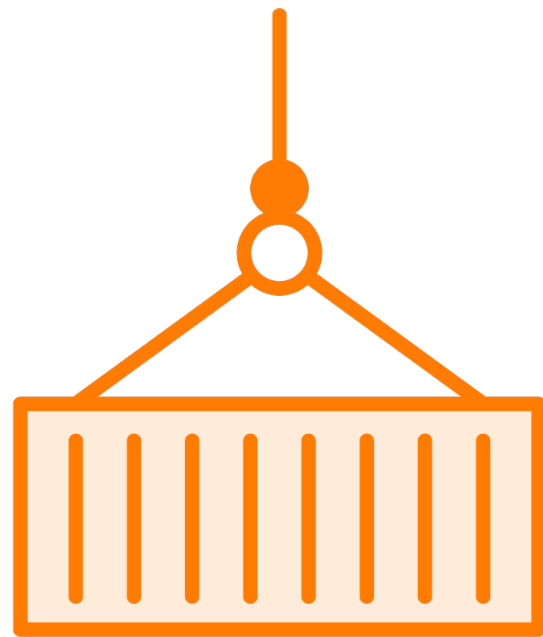
Don't scale well

Most sites use a table-like structure though

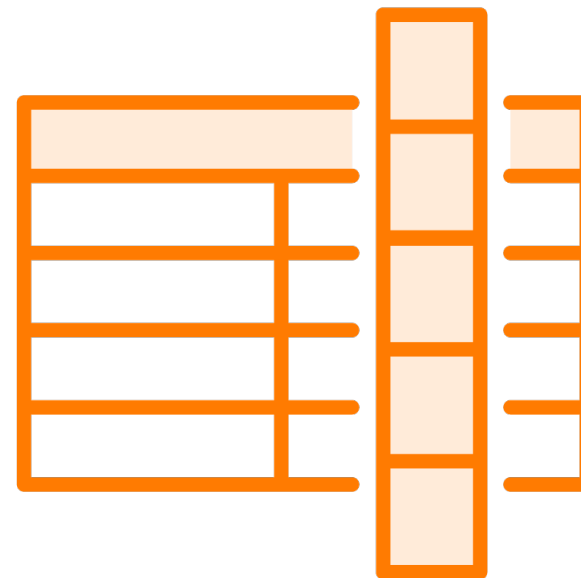
Bootstrap uses Grid



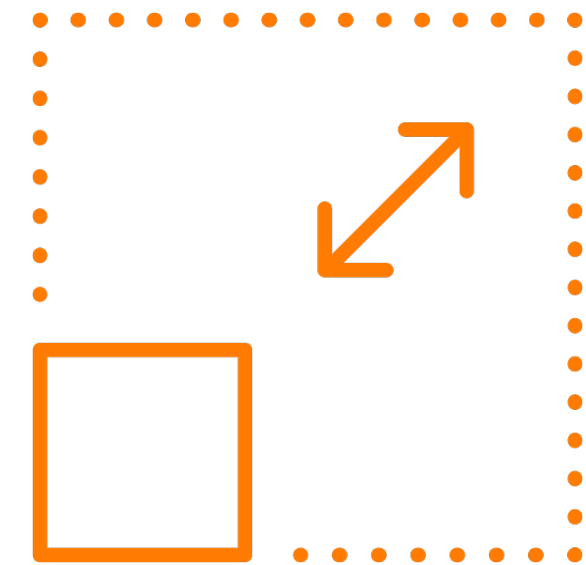
Bootstrap Grid Concepts



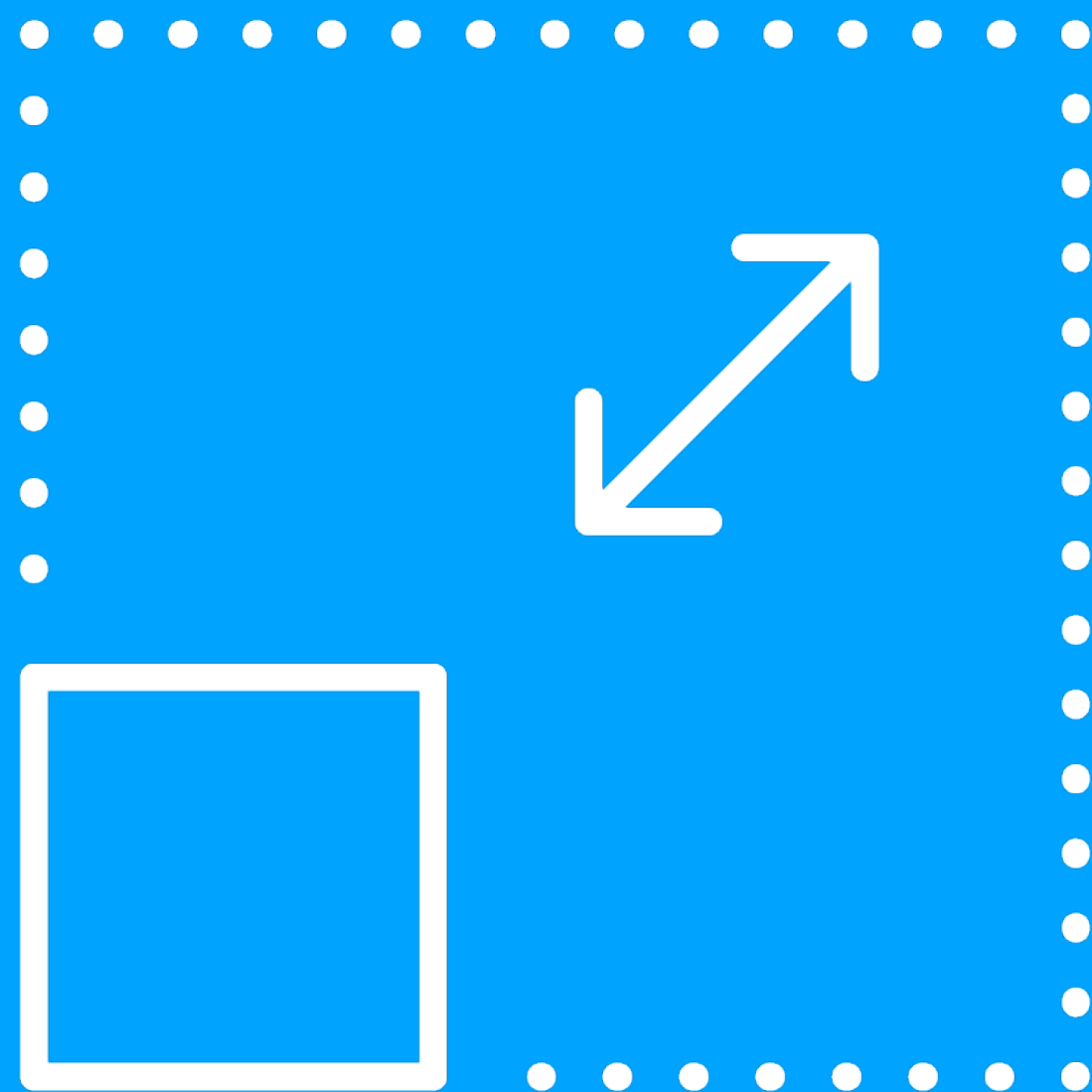
Container



Rows and columns



Breakpoints



Working with Breakpoints

Widths on which layout will “adapt” to different arrangement

Foundation for responsive design

Can be customized, often not needed

Based on media queries



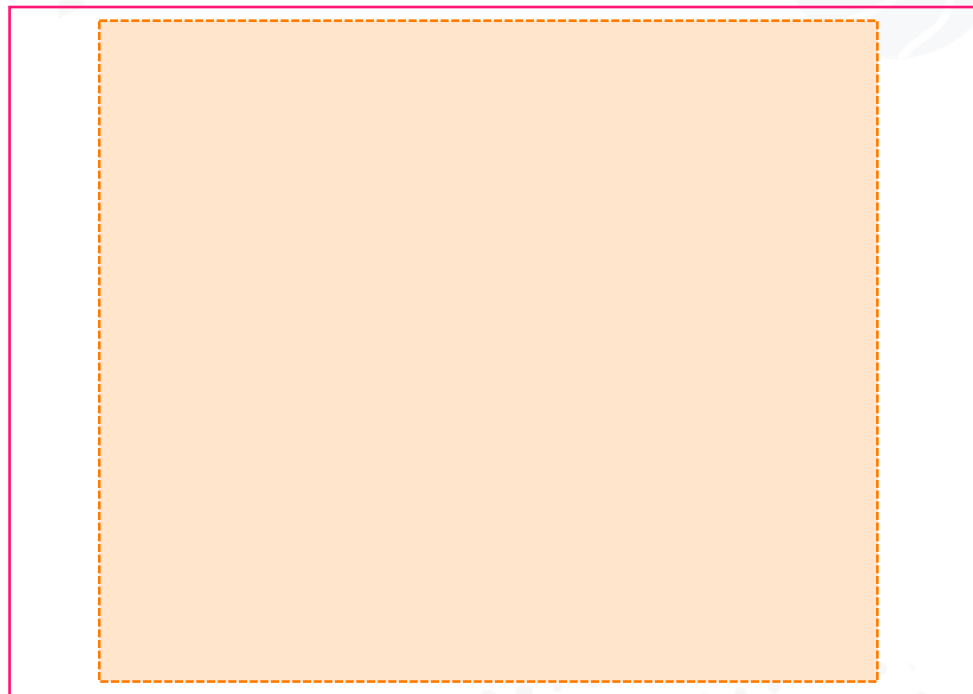
Available Breakpoints

Name	Prefix	Width
Extra small	None	< 576px
Small	sm	>= 576px
Medium	md	>= 768px
Large	lg	>= 992px
Extra large	xl	>= 1200px
Extra extra large	xxl	>= 1400px

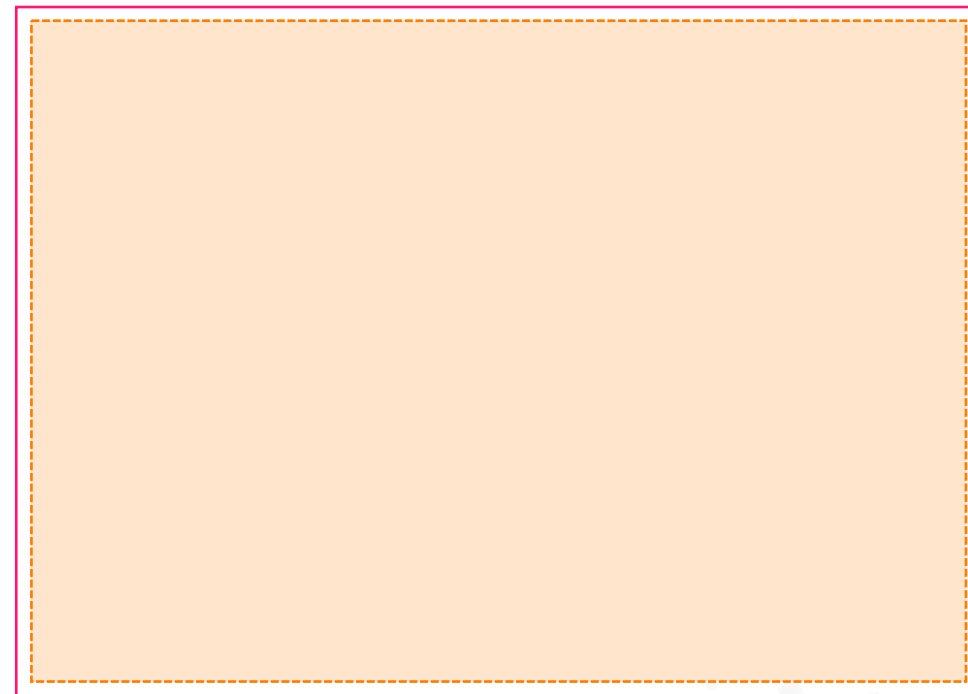


Using the Container

```
<div class="container">  
  ...  
</div>
```



```
<div class="container-fluid">  
  ...  
</div>
```



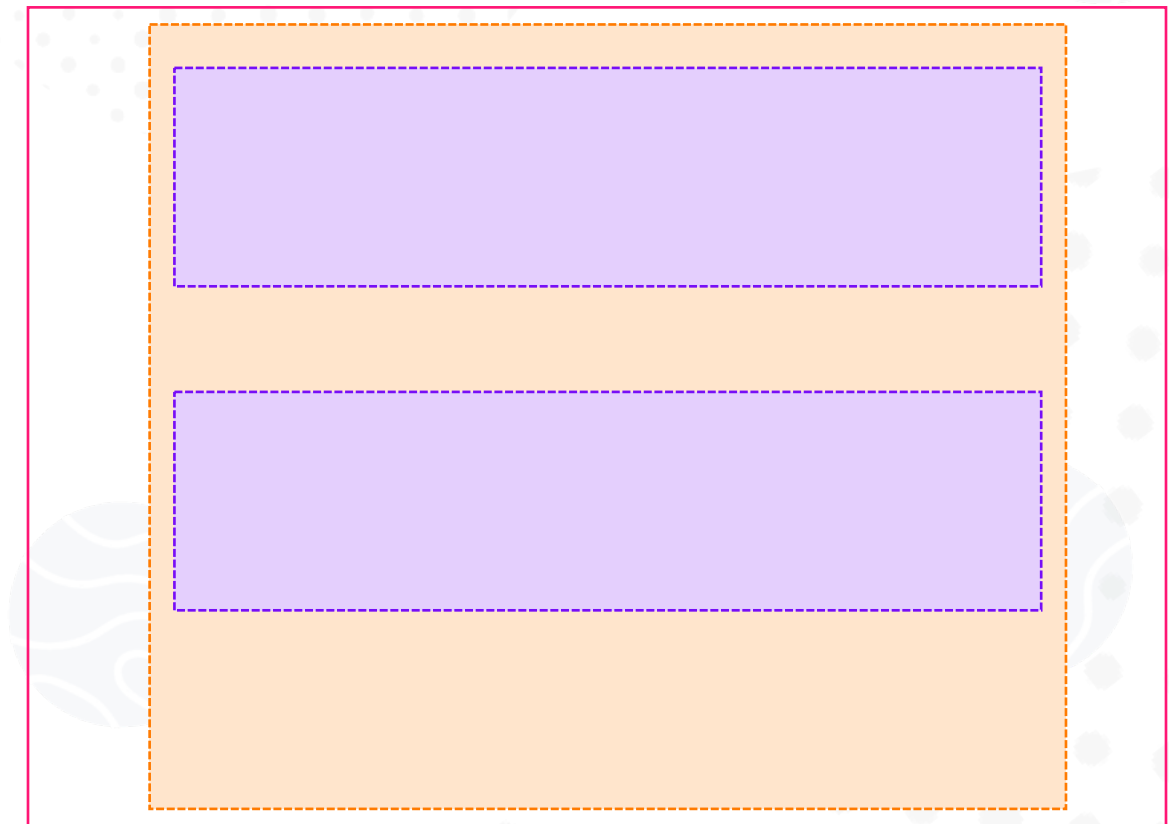
Responsive Containers

Name	Extra small	Small	Medium	Large	X-Large	XX-Large
<code>.container</code>	100%	540px	720px	960px	1140px	1320px
<code>.container-sm</code>	100%	540px	720px	960px	1140px	1320px
<code>.container-md</code>	100%	100%	720px	960px	1140px	1320px
<code>.container-lg</code>	100%	100%	100%	960px	1140px	1320px
<code>.container-xl</code>	100%	100%	100%	100%	1140px	1320px
<code>.container-xxl</code>	100%	100%	100%	100%	100%	1320px
<code>.container-fluid</code>	100%	100%	100%	100%	100%	100%



Adding Rows

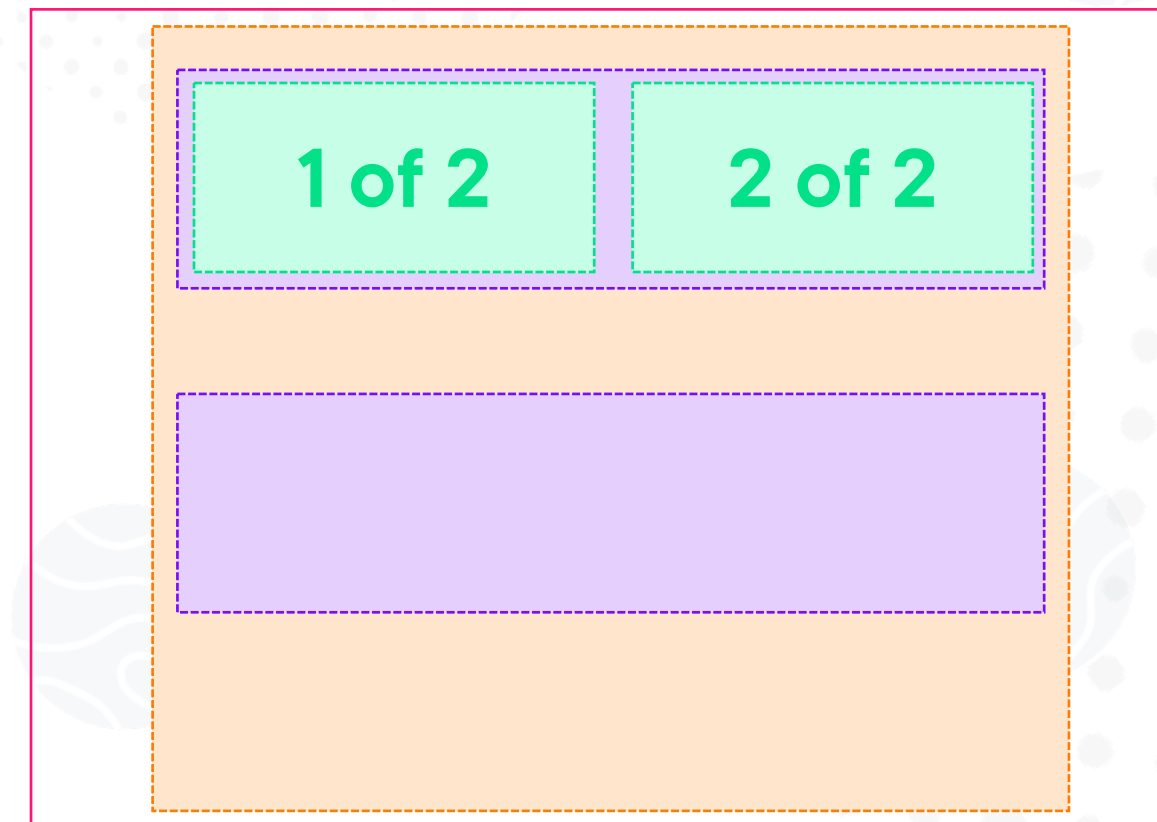
```
<div class="container">  
  <div class="row">  
  </div>  
  
  <div class="row">  
  </div>  
</div>
```

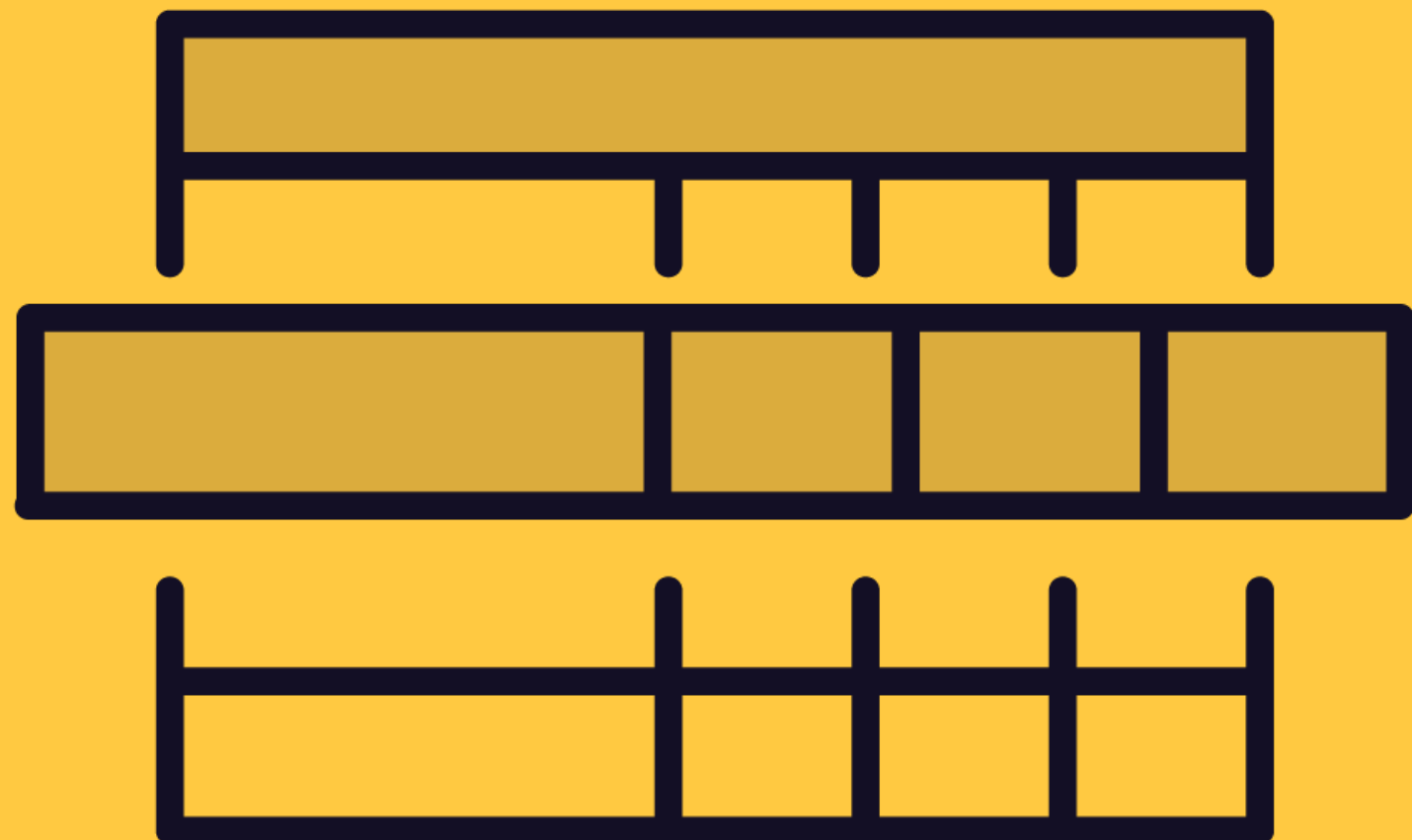


Adding Columns to Rows

```
<div class="container">  
  <div class="row">  
    <div class="col">  
      1 of 2  
    </div>  
    <div class="col">  
      2 of 2  
    </div>  
  </div>  
  <div class="row">  
  </div>  
</div>
```

Columns are equally sized on all sizes





**Each Row Has
12 Columns**

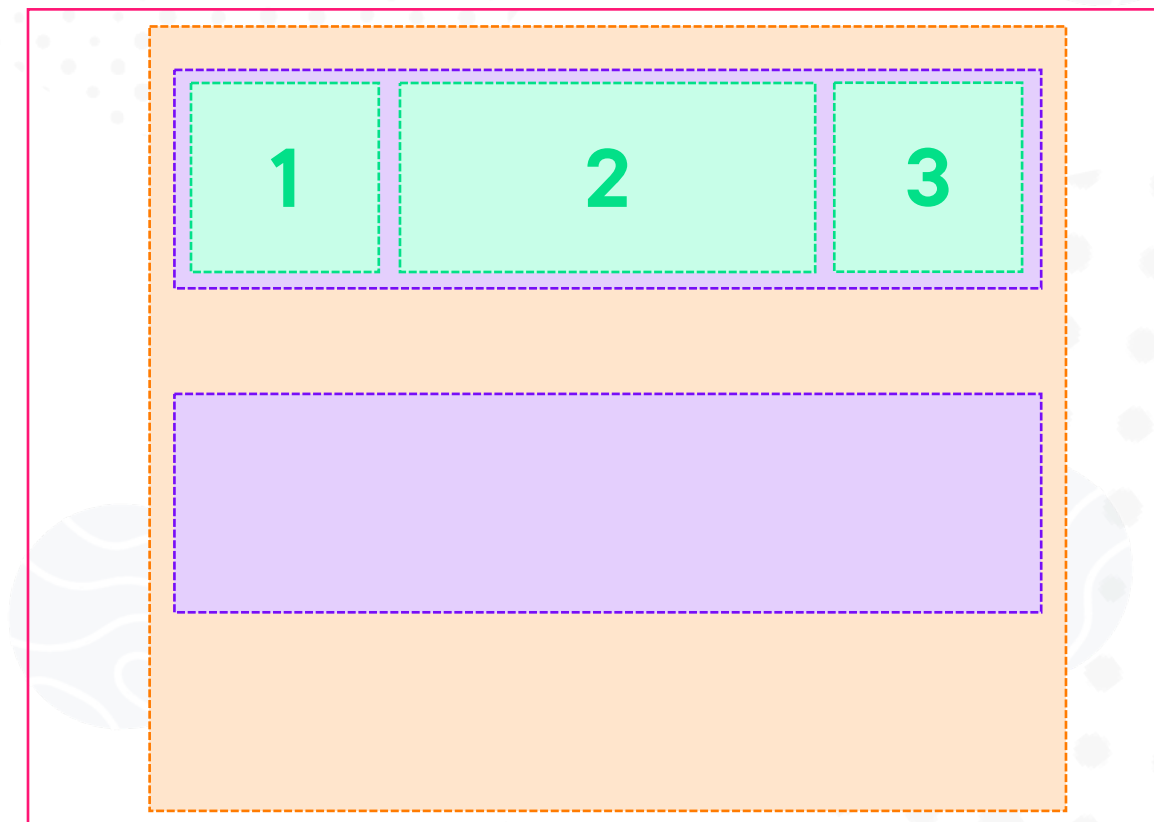
col-xx

Will stack if more are added



Setting One Column Width

```
<div class="container">  
  <div class="row">  
    <div class="col">  
      1  
    </div>  
    <div class="col-6">  
      2  
    </div>  
    <div class="col">  
      3  
    </div>  
  </div>  
  <div class="row">  
  </div>  
</div>
```



Responsiveness Using Breakpoints

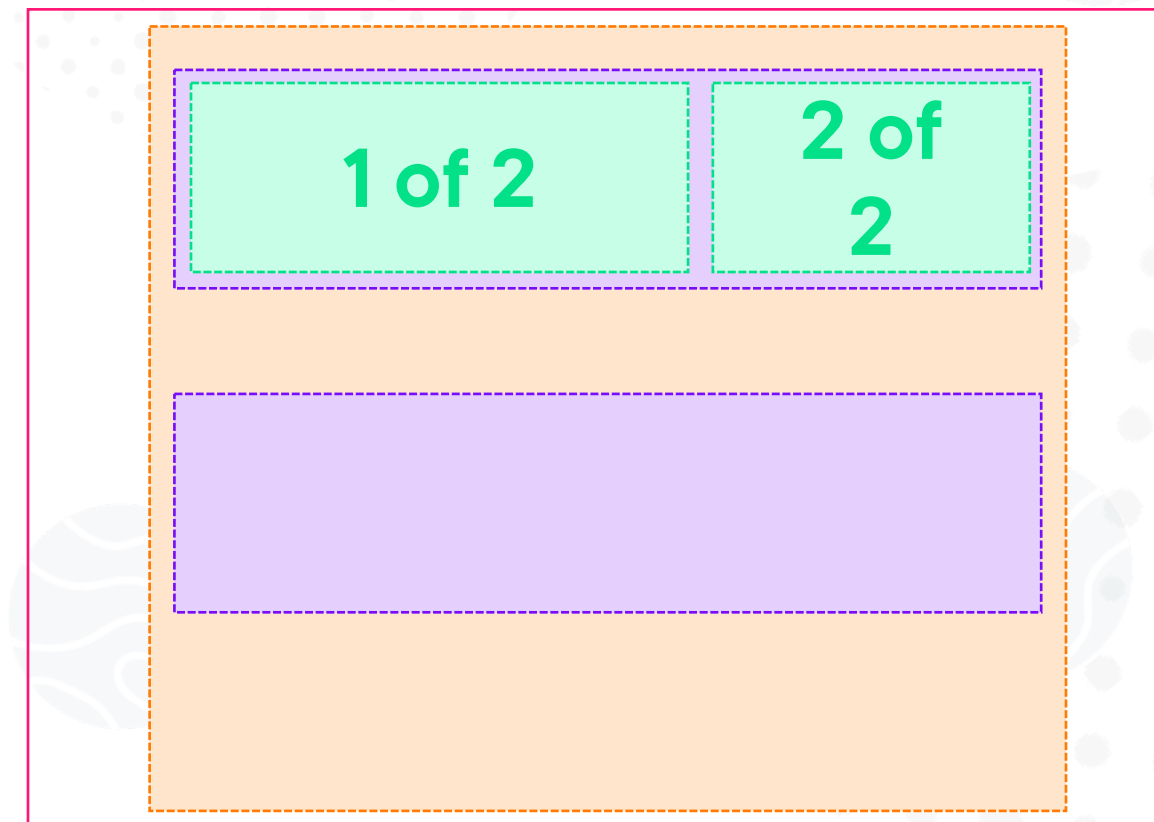
Name	Prefix	Width
Extra small	col-	< 576px
Small	col-sm-	>= 576px
Medium	col-md-	>= 768px
Large	col-lg-	>= 992px
Extra large	col-xl-	>= 1200px
Extra extra large	col-xxl-	>=1400px



Column Naming

```
<div class="container">  
  <div class="row">  
    <div class="col-md-8">  
      1 of 2  
    </div>  
    <div class="col-md-4">  
      2 of 2  
    </div>  
  </div>  
  <div class="row">  
  </div>  
</div>
```

Below md, this will be one column



Column Sizing

		col						col			
--	--	-----	--	--	--	--	--	-----	--	--	--



Column Sizing

		col						col			
	col-4				col-4			col-4			



Column Sizing

		col						col			
	col-4				col-4			col-4			
	col-xx-3		col-xx-3					col-xx-6			



Column Sizing

		col						col		
	col-4			col-4			col-4			
	col-xx-3		col-xx-3				col-xx-6			
col-xx-2	col-xx-2		col-xx-2		col-xx-2		col-xx-2		col-xx-2	



Column Sizing

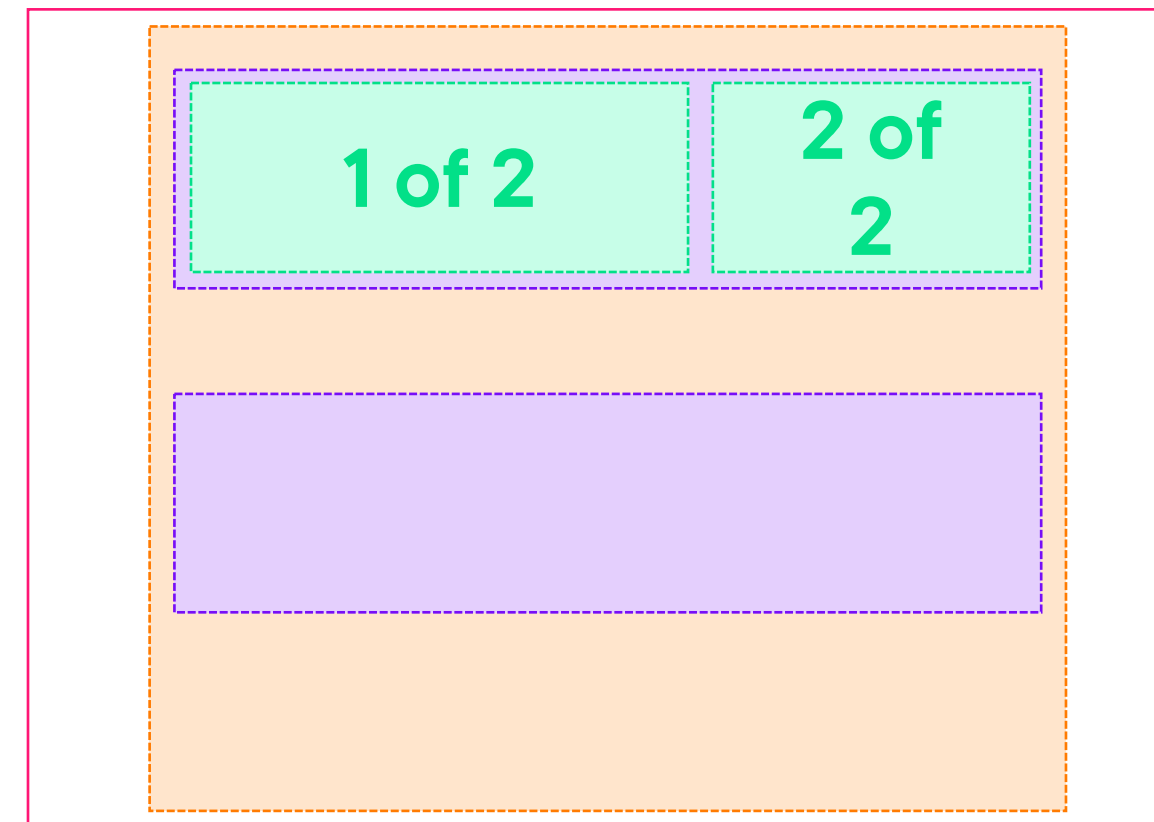
		col					col		
	col-4			col-4			col-4		
	col-xx-3		col-xx-3				col-xx-6		
col-xx-2	col-xx-2		col-xx-2	col-xx-2	col-xx-2	col-xx-2	col-xx-2		
				col-xx-10				col-xx-2	



Combining Column Sizes

```
<div class="row">  
  <div class="col-md-8 col-sm-4">  
    1 of 2  
  </div>  
  <div class="col-md-4 col-sm-8">  
    2 of 2  
  </div>  
</div>
```

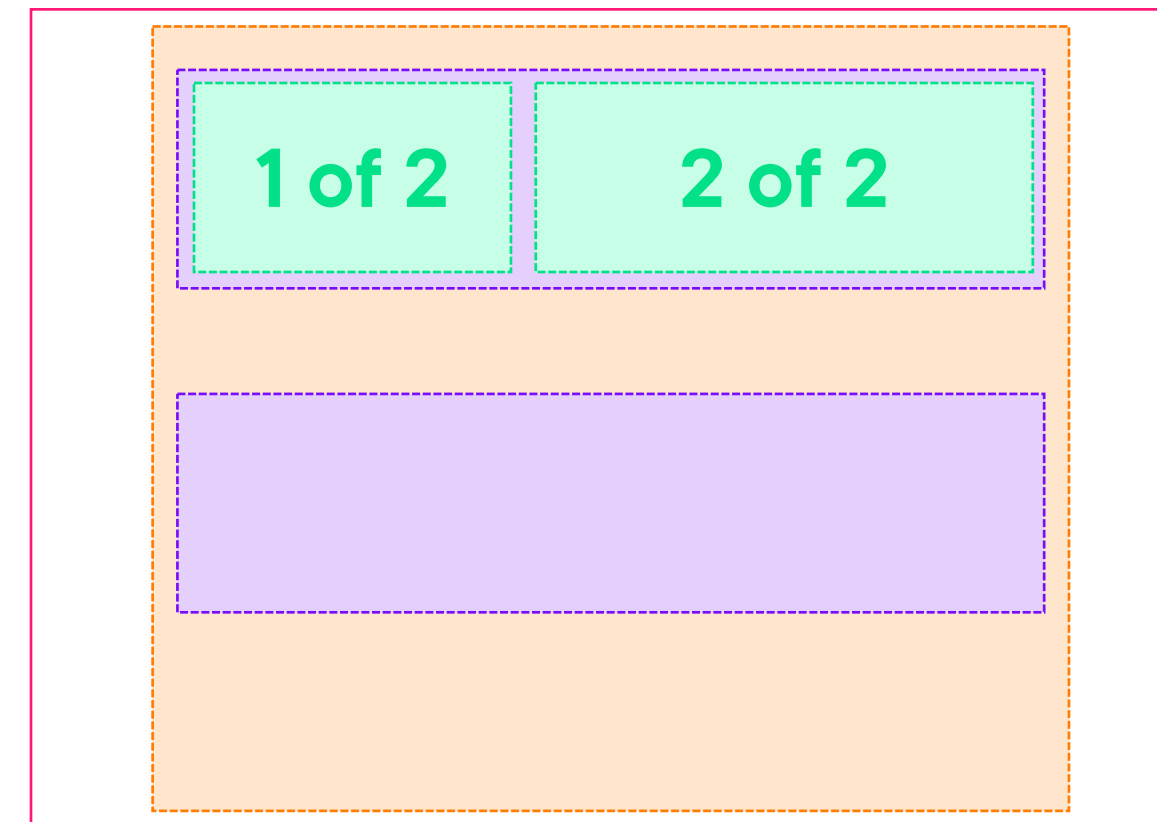
Medium and up



Combining Column Sizes

```
<div class="row">  
  <div class="col-md-8 col-sm-4">  
    1 of 2  
  </div>  
  <div class="col-md-4 col-sm-8">  
    2 of 2  
  </div>  
</div>
```

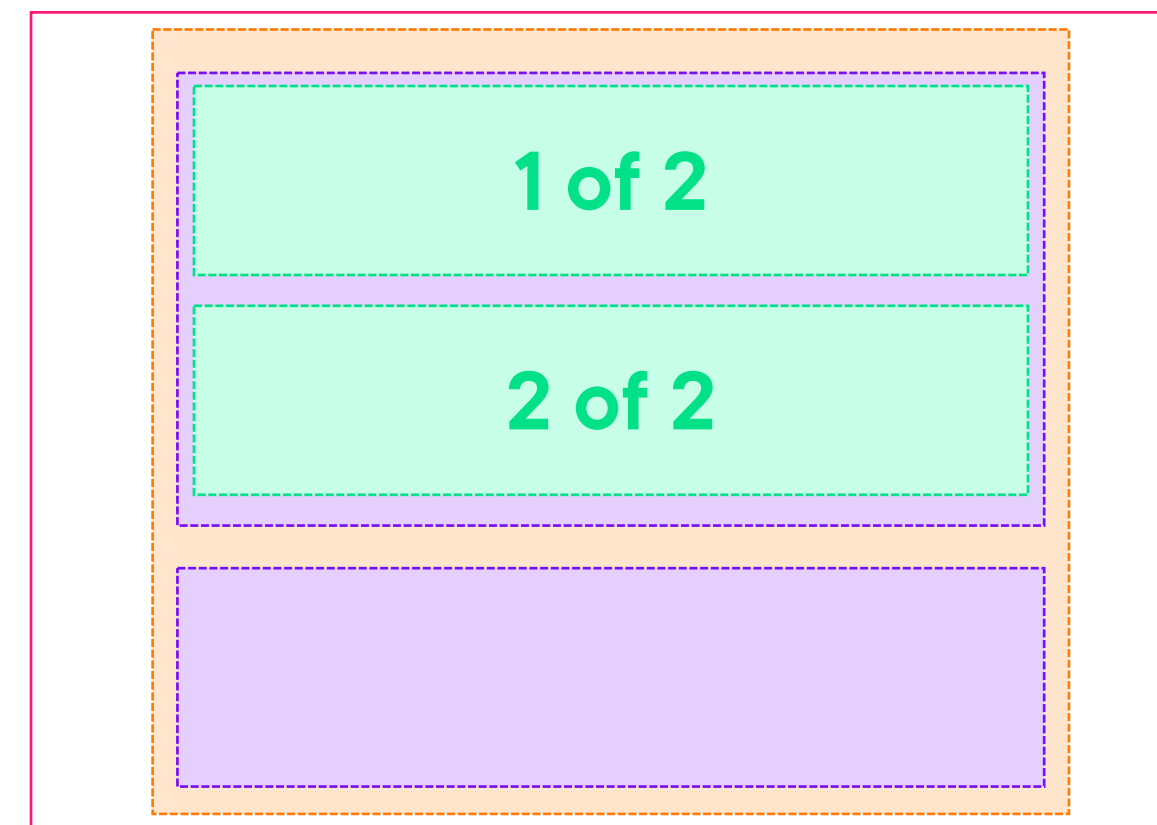
Between small and medium



Combining Column Sizes

```
<div class="row">  
  <div class="col-md-8 col-sm-4">  
    1 of 2  
  </div>  
  <div class="col-md-4 col-sm-8">  
    2 of 2  
  </div>  
</div>
```

Under small

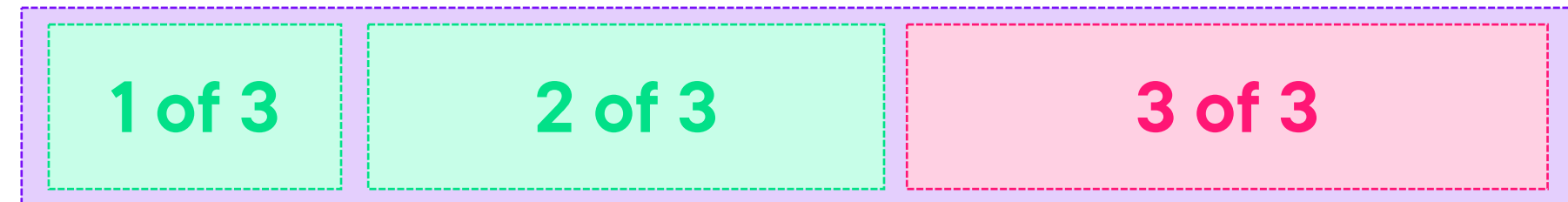


Demo: Understanding the Grid



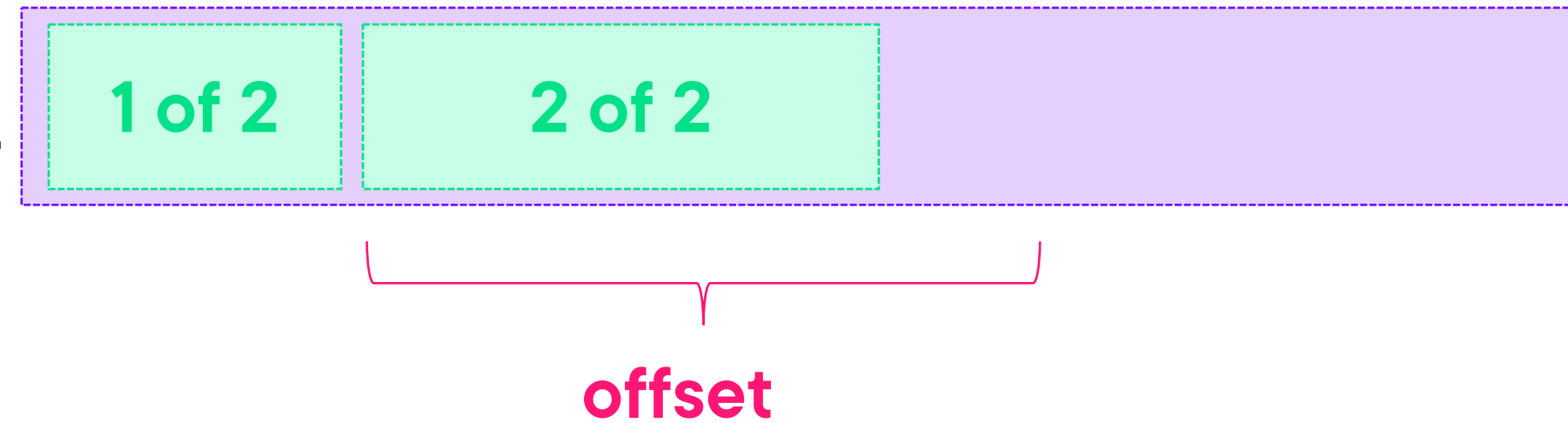
Changing the Order of Columns

```
<div class="row">  
  <div class="col-md-2">  
    1 of 3  
  </div>  
  <div class="col-md-4 order-5 ">  
    2 of 3  
  </div>  
  <div class="col-md-6 order-1 ">  
    3 of 3  
  </div>  
</div>
```



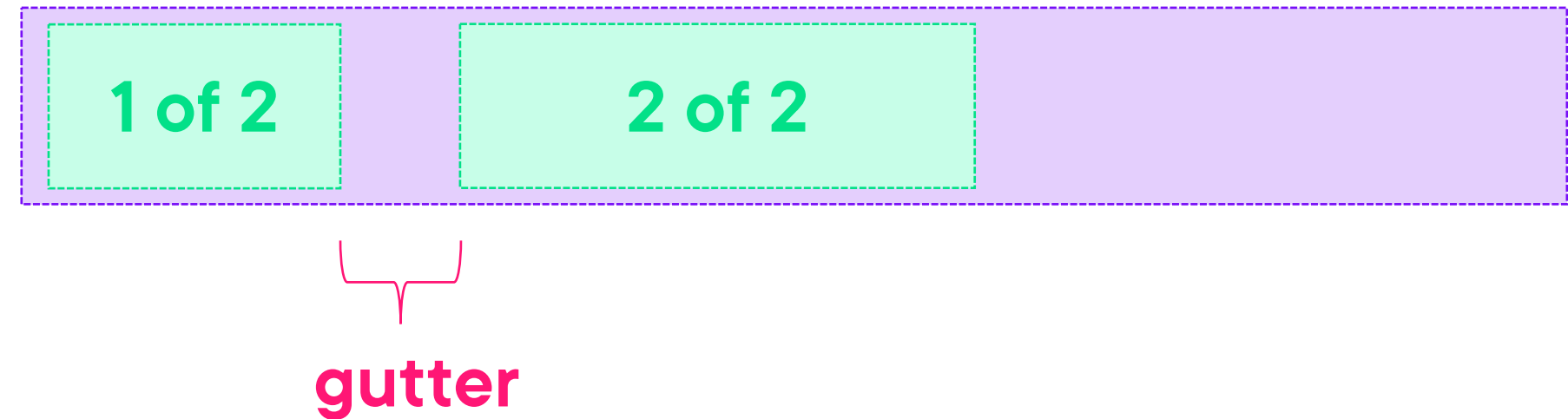
Leaving Some Space

```
<div class="row">  
  <div class="col-md-2">  
    1 of 2  
  </div>  
  <div class="col-md-4 offset-md-6">  
    2 of 2  
  </div>  
</div>
```



Adding Gutters

```
<div class="row gx-4">  
  <div class="col-md-2">  
    <div>1 of 2</div>  
  </div>  
  <div class="col-md-4">  
    <div>2 of 2</div>  
  </div>  
</div>
```



Gutters can also be vertical or both



Demo: Working with Columns



Demo: Making the Detail Page Responsive

